EEG Signal Filtering and Spectral Analysis for Seizure Detection:

Khan, Akif.[1] & Marsellos, A.E.[2]

[1]*Sanford H. Calhoun High School,* 1786 State St, Merrick, NY 11566

[2]*Dept. Geology, Environment and Sustainability, Hofstra University, Hempstead, NY, USA*

Abstract:

Electroencephalography (EEG) provides a noninvasive way to record brain activity, but seizures are difficult to detect because the signals are noisy and often overlap with normal rhythms. In this project, EEG data from seizure and non-seizure cases were analyzed using signal processing methods in R.

Preprocessing included Butterworth and Kolmogorov–Zurbenko (KZ) filtering to reduce noise and enhance meaningful oscillations.

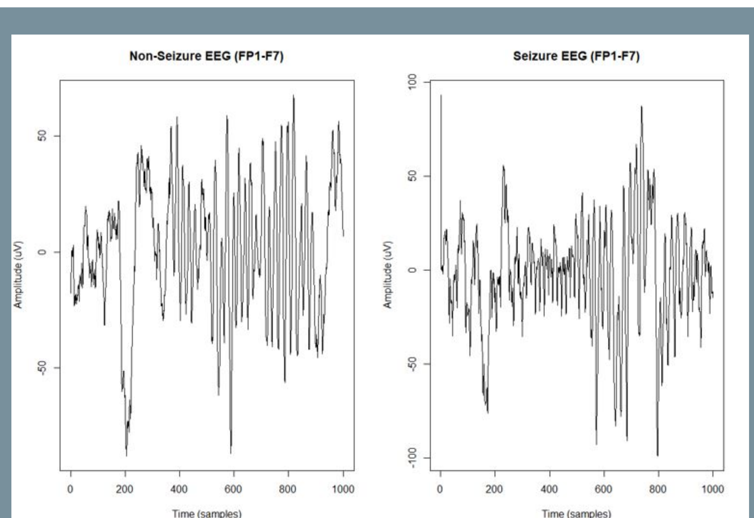
Frequency-domain analysis with Fast Fourier Transform (FFT) and periodograms revealed differences across canonical EEG bands. Seizure signals consistently showed higher delta power and altered alpha activity compared to non-seizure conditions. A systematic ranking of channels confirmed strong shifts in band power during seizures, especially in delta increases and alpha suppression.

These findings demonstrate that filtering and spectral analysis can highlight seizure-related brain activity and provide a foundation for building automated seizure detection tools that could improve diagnosis and real-time monitoring.

Background Information:

- Epileptic seizures often occur without warning, creating risks for patients.
- EEG measures brainwave activity, but seizure patterns are subtle and buried in noise.
- Signal detection and filtering are critical to reveal useful features.
- Previous studies suggest frequency bands (delta, theta, alpha, beta, gamma) hold important clues for seizure onset.

Figure 1: Raw EEG signals for non-seizure (left) and seizure (right), FP1–F7 channel.



Method/Materials:

- **Data**: EEG recordings from seizure (chb01_03) and non-seizure (chb01_04) datasets.
- **Preprocessing**:
  - Applied Butterworth filter and KZ filter for noise removal.
  - Extracted FP1–F7 and other shared channels.
- **Spectral Analysis**:
  - Fast Fourier Transform (FFT) to capture frequency content.
  - Periodogram (power spectral density estimation) for seizure vs non-seizure comparison.
- **Band Power Extraction**:
  - Calculated relative power in delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (30–50 Hz).
  - Normalized values and ranked channels by seizure vs non-seizure differences.
- **Software**: RStudio (v4.2.1) with packages `kza`, `TSA`, `imputeTS`, and base spectral functions.

Results:

Time Domain (Fig. 1 & Fig. 2): Raw seizure EEG showed noisy fluctuations and sudden amplitude spikes. Filtering reduced noise and clarified underlying rhythms.

FFT Analysis (Fig. 3): Seizure signals displayed distinct peaks in lower frequencies, while non-seizure activity had more distributed power.

Periodogram Comparison (Fig. 4): Overlay plots of seizure vs non-seizure revealed increased low-frequency power during seizures, as well as altered complexity measured by entropy.

Band Power Analysis (Fig. 5): Delta power consistently increased during seizures across multiple channels, while alpha power decreased. Theta and gamma bands showed moderate but noticeable differences.

Figure 2: Filtered EEG signals for non-seizure (left) and seizure (right), FP1–F7 channel.

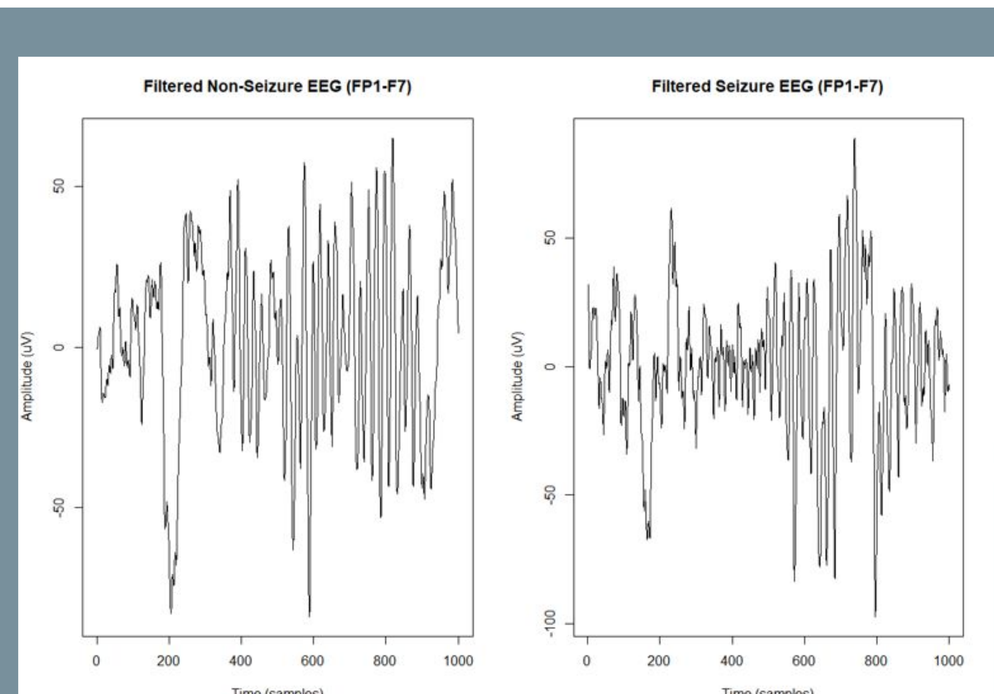


Figure 3: FFT plots for seizure (red) and non-seizure (blue), showing differences in spectral peaks.

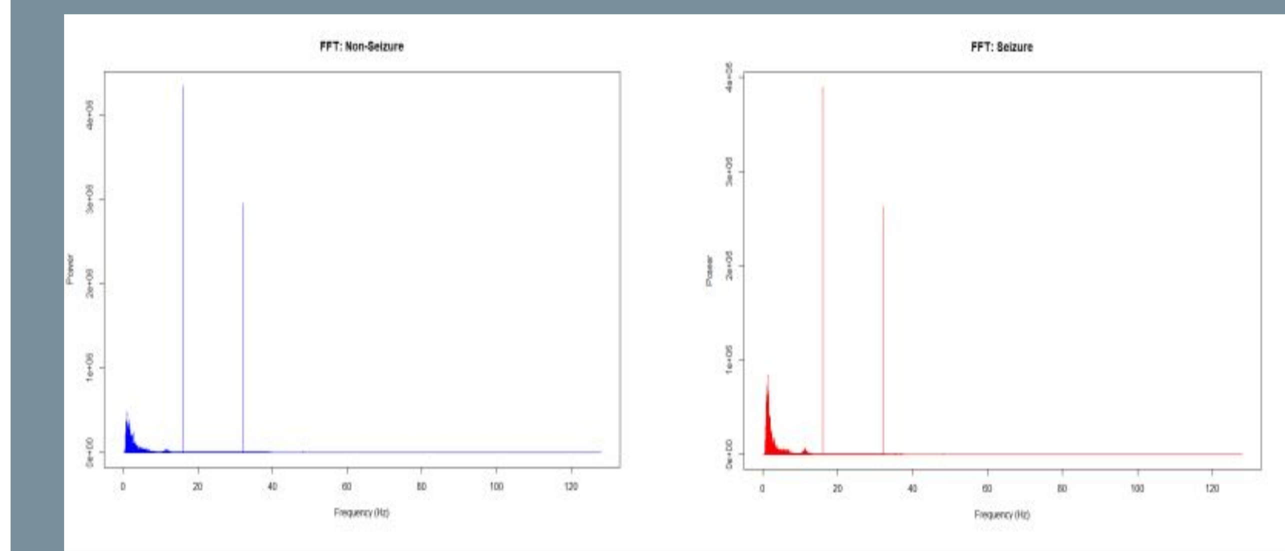


Figure 4: Periodogram comparison (P7–T7 channel), highlighting seizure-related increases in low-frequency power.

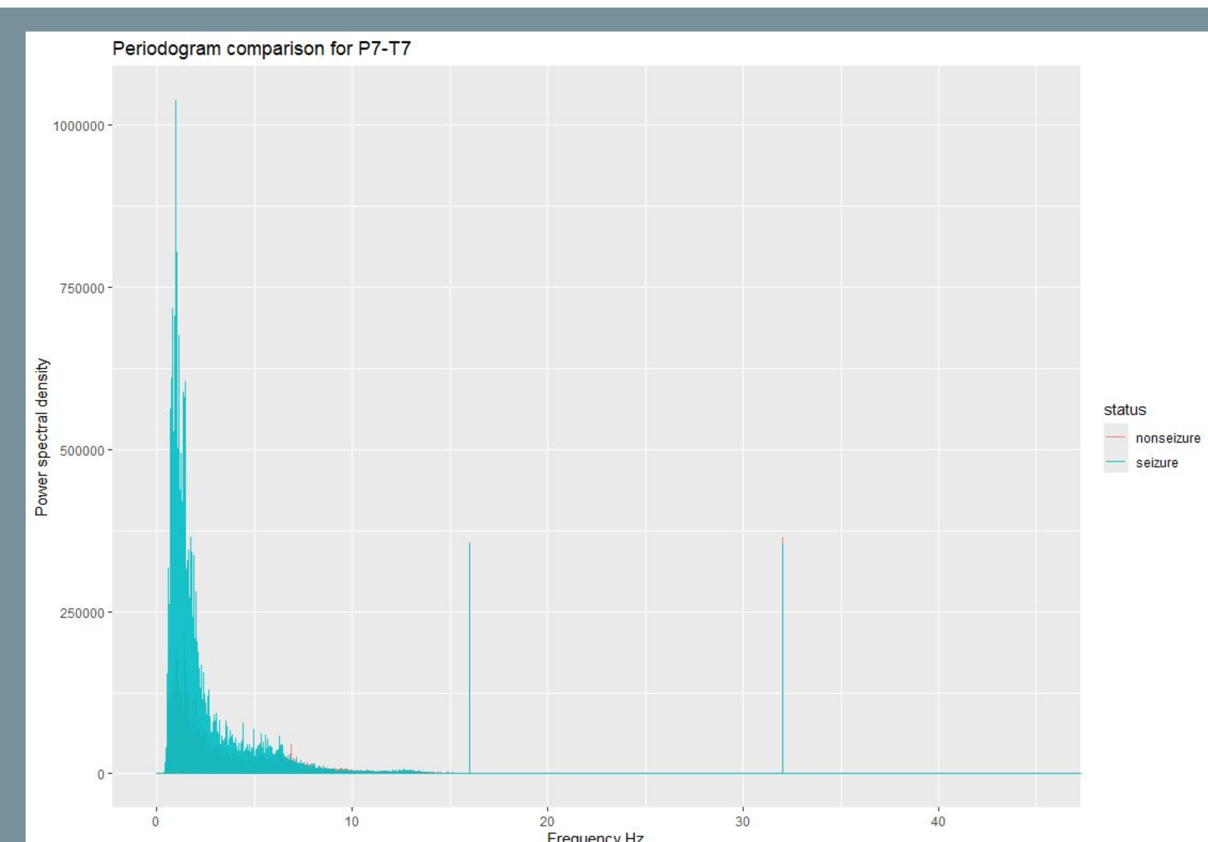
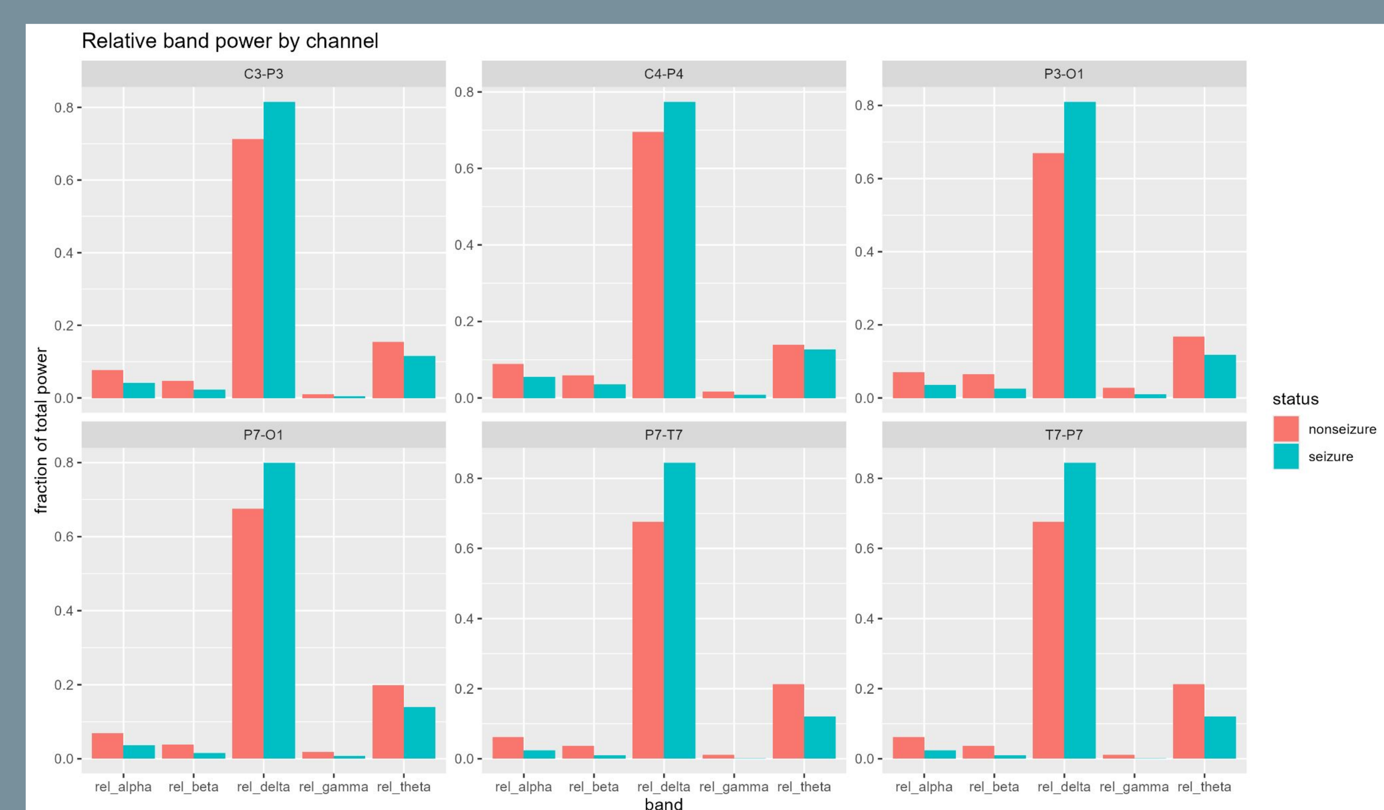


Figure 5: Relative band power across EEG channels, showing increased delta and altered alpha during seizures.



Discussion/Conclusion:

- Filtering combined with spectral and band power analysis provides a reliable workflow for highlighting seizure-related EEG features.
- These methods strengthen the case for developing automated detection systems that could support neurologists and improve patient safety.
- Future work will include:
  - multi-channel comparisons
  - entropy-based classifiers
  - integration with machine learning approaches.

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